

Boosted Gaussian Witness Kick – A Manufactured Solution

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1 Purpose

This note defines a manufactured solution for a low-energy witness e-/e+ pair, kicked by the electromagnetic field of two highly relativistic and colliding Gaussian electron bunches. The goal is to guide an OPALX implementation and establish a order of magnitude estimate.

It is understood that for the detector design, of a γ - γ collider, a full self-consistent model is essential.

Each source bunch is prescribed as a uniformly (non accelerated) moving, spherical Gaussian charge distribution in its own rest frame. The rest-frame Coulomb field is known analytically and is Lorentz-transformed to the lab frame at the witness event. The resulting lab-frame field is then applied to one witness electron with the same Boris kick convention used by OPALX.

2 Current OPALX Witness Model

The present OPALX witness treatment is a source-frame field sampling model. The source bunch defines the mesh and the field solve. A witness particle is not deposited on that mesh; instead its position is translated and rotated into the source beam-beam frame, the already available mesh field is gathered there, and the gathered vectors are rotated back before applying the usual particle kick.

In this approximation the witness is represented only by its spatial coordinate in the active tracking step. The relevant mapping is geometric,

$$\mathbf{r}_s = R_s (\mathbf{r}_w + (s_w - s_s)\hat{\mathbf{z}}),$$

where s_s and s_w are the source and witness path lengths and R_s is the rigid rotation into the source beam frame. The sampled fields are then

$$\mathbf{E}_{\text{used}} = R_s^T \mathbf{E}_s(\mathbf{r}_s), \quad \mathbf{B}_{\text{used}} = R_s^T \mathbf{B}_s(\mathbf{r}_s).$$

This is useful as a first implementation because it reuses the **BeamBeam** mesh, but it does not treat the field evaluation as a Lorentz-covariant event calculation.

The full relativistic reference model will eventually starts from the witness event

$$x_w^\mu = (ct_w, \mathbf{x}_w)$$

and evaluates the contribution of each source bunch j at that event. Let the source move with almost constant **BeamBeam** velocity

$$\mathbf{v}_j = c\boldsymbol{\beta}_j, \quad \gamma_j = \frac{1}{\sqrt{1 - \boldsymbol{\beta}_j^2}},$$

and define the event relative to the source centroid by

$$\Delta t_j = t_w - t_{j0}, \quad \Delta \mathbf{x}_j = \mathbf{x}_w - \mathbf{X}_{j0}.$$

With the decomposition into components parallel and transverse to β_j , the Lorentz transformation into the source rest frame is

$$ct'_j = \gamma_j (c\Delta t_j - \beta_j \cdot \Delta \mathbf{x}_j),$$

$$\Delta \mathbf{x}'_{j,\parallel} = \gamma_j (\Delta \mathbf{x}_{j,\parallel} - c\beta_j \Delta t_j), \quad \Delta \mathbf{x}'_{j,\perp} = \Delta \mathbf{x}_{j,\perp}.$$

The term $\beta_j \cdot \Delta \mathbf{x}_j / c$ is the relativity-of-simultaneity piece that is absent in the current geometric sampling model.

For the manufactured Gaussian reference used here, the source rest-frame charge density is

$$\rho'_j(r') = \frac{Q_j}{(2\pi)^{3/2}\sigma_j^3} \exp\left(-\frac{r'^2}{2\sigma_j^2}\right),$$

and the electrostatic field is

$$\mathbf{E}'_j(\mathbf{r}') = \frac{Q_j}{4\pi\epsilon_0} \frac{\mathbf{r}'}{r'^3} \left[\operatorname{erf}\left(\frac{r'}{\sqrt{2}\sigma_j}\right) - \sqrt{\frac{2}{\pi}} \frac{r'}{\sigma_j} \exp\left(-\frac{r'^2}{2\sigma_j^2}\right) \right], \quad \mathbf{B}'_j = 0.$$

More generally, the source rest-frame field tensor is transformed back to the lab/reference frame,

$$F_{j,\text{lab}}^{\mu\nu}(x_w) = \Lambda^\mu{}_\alpha(\beta_j) F_j^{\prime\alpha\beta}(x'_j) \Lambda^\nu{}_\beta(\beta_j).$$

In vector form this is

$$\mathbf{E}_j = \gamma_j (\mathbf{E}'_j - \mathbf{v}_j \times \mathbf{B}'_j) - \frac{\gamma_j^2}{\gamma_j + 1} \frac{\mathbf{v}_j}{c^2} (\mathbf{v}_j \cdot \mathbf{E}'_j),$$

$$\mathbf{B}_j = \gamma_j \left(\mathbf{B}'_j + \frac{\mathbf{v}_j \times \mathbf{E}'_j}{c^2} \right) - \frac{\gamma_j^2}{\gamma_j + 1} \frac{\mathbf{v}_j}{c^2} (\mathbf{v}_j \cdot \mathbf{B}'_j).$$

The fields acting on the witness are the superposition

$$\mathbf{E}(x_w) = \sum_j \mathbf{E}_j(x_w), \quad \mathbf{B}(x_w) = \sum_j \mathbf{B}_j(x_w).$$

For the OPALX normalized momentum $\mathbf{P} = \gamma_w \beta_w$, the corresponding witness equations of motion are

$$\frac{d\mathbf{x}_w}{dt} = c \frac{\mathbf{P}}{\sqrt{1 + \mathbf{P}^2}}, \quad \frac{d\mathbf{P}}{dt} = \frac{q}{mc} \left[\mathbf{E}(x_w) + c \frac{\mathbf{P}}{\sqrt{1 + \mathbf{P}^2}} \times \mathbf{B}(x_w) \right].$$

This is the model the manufactured solution is meant to test against: event transformation into each source rest frame, field evaluation there, field-tensor transformation back to the lab/reference frame, and then the witness kick.

During an active **BEAMBEAM** window, OPALX treats particle container 0 as the source bunch. Additional containers listed through **WITNESS_CONTAINERS** are passive: their charge is not deposited onto the source mesh, and they only sample the field that was already computed from container 0.

Let the source container path length be s_s , the witness container path length be s_w , and the witness particle coordinate in its own container frame be

$$\mathbf{r}_w = (x_w, y_w, z_w).$$

The implemented longitudinal mapping into the source frame is

$$\Delta s = s_w - s_s, \quad z_s = z_w + \Delta s.$$

With R_s the rigid rotation from the reference frame into the source beam frame, the sampled source-frame coordinate is

$$\mathbf{r}_s = R_s (\mathbf{r}_w + \Delta s \hat{\mathbf{z}}).$$

OPALX then gathers the current source mesh fields at this point:

$$\mathbf{E}_s(\mathbf{r}_s), \quad \mathbf{B}_s(\mathbf{r}_s).$$

After gathering, the witness particle positions are restored and the fields are rotated back with the inverse rigid rotation,

$$\mathbf{E}_{\text{used}} = R_s^T \mathbf{E}_s, \quad \mathbf{B}_{\text{used}} = R_s^T \mathbf{B}_s.$$

Those fields are passed to the normal per-container Boris kick. In OPALX units the particle momentum is

$$\mathbf{P} = \beta_w \gamma_w,$$

and the continuous Lorentz-force equation represented by the kick is

$$\frac{d\mathbf{P}}{dt} = \frac{qc}{mc^2} \mathbf{E} + \frac{qc^2}{mc^2} \beta_w \times \mathbf{B}.$$

2.1 Source Field Currently Available in OPALX

The `BeamBeam` field calculation uses the binned quasi-static field path. This statement includes the case where the adaptive binning has only one current bin: as long as a `BINNING` definition is attached to the field solver, OPALX dispatches to `computeBinnedSelfFields()` and applies the same quasi-static Lorentz correction to that single bin.

For a source bin b , OPALX computes the global mean normalized momentum

$$\bar{\mathbf{P}}_b = \langle \beta \gamma \rangle_b,$$

and the bin Lorentz factor and mean velocity

$$\gamma_b = \sqrt{1 + \bar{\mathbf{P}}_b^2}, \quad \mathbf{v}_b = c \frac{\bar{\mathbf{P}}_b}{\gamma_b}.$$

Before the Poisson solve, the mesh density is transformed to the approximate bin rest frame by

$$\rho'_b = \frac{\rho_b}{\gamma_b},$$

and the longitudinal mesh spacing is stretched by

$$\Delta z' = \gamma_b \Delta z.$$

After solving the electrostatic problem in this stretched frame, the code accumulates the lab-frame fields

$$\begin{aligned} \mathbf{E}_b &= \gamma_b \mathbf{E}'_b - (\gamma_b - 1)(\mathbf{E}'_b \cdot \hat{\mathbf{w}}_b) \hat{\mathbf{w}}_b, \\ \mathbf{B}_b &= \frac{\gamma_b}{c^2} \mathbf{v}_b \times \mathbf{E}'_b, \quad \hat{\mathbf{w}}_b = \frac{\mathbf{v}_b}{|\mathbf{v}_b|}. \end{aligned}$$

The accumulated fields are then gathered to source particles, and the current mesh fields can also be sampled by witness containers.

2.2 What Is Missing

The witness model is still a source-frame field sampling approximation. It rotates and translates witness geometry into the source beam frame, samples the mesh, and rotates the sampled vectors back. It does not evaluate the electromagnetic field tensor at the witness event.

A covariant treatment would start from a witness event

$$x_w^\mu = (ct_w, \mathbf{x}_w)$$

and transform that event into the rest frame of each source bunch. For a source with velocity $\mathbf{v}_s = c\boldsymbol{\beta}_s$, the parallel part of the event transform is

$$t'_s = \gamma_s \left(t_w - \frac{\boldsymbol{\beta}_s \cdot \mathbf{x}_w}{c} \right),$$

$$\mathbf{x}'_{s,\parallel} = \gamma_s (\mathbf{x}_{w,\parallel} - \mathbf{v}_s t_w), \quad \mathbf{x}'_{s,\perp} = \mathbf{x}_{w,\perp}.$$

The field tensor would then be transformed back to the lab/reference frame,

$$F_{\text{lab}}^{\mu\nu} = \Lambda^\mu{}_\alpha F_s'^{\alpha\beta} \Lambda^\nu{}_\beta.$$

In vector notation, for a source rest-frame field $\mathbf{E}'$, $\mathbf{B}'$, the lab field is

$$\begin{aligned} \mathbf{E} &= \gamma_s (\mathbf{E}' - \mathbf{v}_s \times \mathbf{B}') - \frac{\gamma_s^2}{\gamma_s + 1} \frac{\mathbf{v}_s}{c^2} (\mathbf{v}_s \cdot \mathbf{E}'), \\ \mathbf{B} &= \gamma_s \left(\mathbf{B}' + \frac{\mathbf{v}_s \times \mathbf{E}'}{c^2} \right) - \frac{\gamma_s^2}{\gamma_s + 1} \frac{\mathbf{v}_s}{c^2} (\mathbf{v}_s \cdot \mathbf{B}'). \end{aligned}$$

The current witness gather does not perform this per-source or per-witness event transformation.

The missing physics and numerical pieces are:

- no Lorentz transformation of the witness event into each source frame;
- no relativity-of-simultaneity term in the sampled source coordinate;
- no full $F^{\mu\nu}$ transformation for the sampled field;
- no retarded-time or Lienard-Wiechert treatment for accelerated sources;
- no deposition of witness charge back onto the source mesh;
- no witness-witness or electron-positron mutual space charge;
- no separate Lorentz transformation of the already accumulated source mesh fields into the low-energy witness frame.

3 Manufactured Boosted-Gaussian Model

For a controlled implementation target, prescribe each source bunch j by

$$Q_j, \quad \sigma'_j, \quad \beta_j, \quad \gamma_j = \frac{1}{\sqrt{1 - \beta_j^2}},$$

and by the lab-frame center worldline

$$\mathbf{R}_j(t) = \mathbf{R}_{j0} + \boldsymbol{\beta}_j c(t - t_0).$$

At the witness event $(t, \mathbf{r}_w)$, define the lab-frame displacement from the source center,

$$\boldsymbol{\ell}_j = \mathbf{r}_w - \mathbf{R}_j(t).$$

Split it into parts parallel and perpendicular to the source velocity,

$$\boldsymbol{\ell}_{j,\parallel} = (\boldsymbol{\ell}_j \cdot \hat{\boldsymbol{\beta}}_j) \hat{\boldsymbol{\beta}}_j, \quad \boldsymbol{\ell}_{j,\perp} = \boldsymbol{\ell}_j - \boldsymbol{\ell}_{j,\parallel}.$$

For a uniformly moving source, the corresponding source-rest coordinate at the same lab event is

$$\mathbf{r}'_j = \boldsymbol{\ell}_{j,\perp} + \gamma_j \boldsymbol{\ell}_{j,\parallel}.$$

The rest-frame charge density is a spherical Gaussian,

$$\rho'_j(\mathbf{r}') = \frac{Q_j}{(2\pi)^{3/2} \sigma_j'^3} \exp\left(-\frac{r'^2}{2\sigma_j'^2}\right).$$

The analytic rest-frame electrostatic potential and electric field are

$$\phi'_j(\mathbf{r}') = \frac{Q_j}{4\pi\epsilon_0} \frac{\text{erf}(u)}{r'}, \quad u = \frac{r'}{\sqrt{2}\sigma_j'},$$

$$\mathbf{E}'_j(\mathbf{r}') = \frac{Q_j}{4\pi\epsilon_0} \frac{\text{erf}(u) - \sqrt{2/\pi} (r'/\sigma_j') \exp[-r'^2/(2\sigma_j'^2)]}{r'^3} \mathbf{r}',$$

with the regular limit $\mathbf{E}'_j(0) = 0$. Since the source is electrostatic in its own rest frame,

$$\mathbf{B}'_j = 0.$$

Boosting this field into the lab gives the simpler component form

$$\mathbf{E}_{j,\parallel} = \mathbf{E}'_{j,\parallel}, \quad \mathbf{E}_{j,\perp} = \gamma_j \mathbf{E}'_{j,\perp},$$

$$\mathbf{B}_j = \frac{1}{c^2} \mathbf{v}_j \times \mathbf{E}_j.$$

The total manufactured lab field is

$$\mathbf{E} = \sum_j \mathbf{E}_j, \quad \mathbf{B} = \sum_j \mathbf{B}_j.$$

3.1 Witness Kick

The manufactured witness uses the same normalized momentum convention as OPALX,

$$\mathbf{P} = \boldsymbol{\beta}_w \gamma_w.$$

The Python prototype applies the same Boris kick sequence used in OPALX:

$$\mathbf{P}^- = \mathbf{P}^n + \frac{1}{2} \Delta t \frac{qc}{mc^2} \mathbf{E}, \tag{1}$$

$$\mathbf{t} = \frac{1}{2} \Delta t \frac{qc^2}{\gamma^- mc^2} \mathbf{B}, \quad \mathbf{s} = \frac{2\mathbf{t}}{1 + \mathbf{t} \cdot \mathbf{t}}, \tag{2}$$

$$\mathbf{P}' = \mathbf{P}^- + \mathbf{P}^- \times \mathbf{t} \tag{3}$$

$$\mathbf{P}^+ = \mathbf{P}' + \mathbf{P}' \times \mathbf{s}, \tag{4}$$

$$\mathbf{P}^{n+1} = \mathbf{P}^+ + \frac{1}{2} \Delta t \frac{qc}{mc^2} \mathbf{E}. \tag{5}$$

3.2 Initial Manufactured Cases

The first Python prototype is

`sandbox/python/boosted_gaussian_witness.py.`

The table below was generated with

$$\begin{aligned} K_s &= 245 \text{ MeV}, \\ \gamma_s &= 480.453039976, \\ \beta_s &= 0.999997833949, \\ Q &= -2.002721 \text{ nC}, \\ \sigma' &= 1 \text{ mm}, \\ \Delta t &= 1 \text{ ps}, \\ \mathbf{P}_w^0 &= (0, 0, 0). \end{aligned}$$

The two symmetric source centers are both at the IP at $t = 0$, moving in opposite z -directions. The witness charge is an electron charge.

case	$\mathbf{r}_w$ [mm]	$ \mathbf{E} $ [GV/m]	$ \mathbf{B} $ [T]	$\Delta \mathbf{P}$
symmetric center	(0.000, 0.000, 0.000)	0.000	0.000	(0.000, 0.000, 0.000)
symmetric off-axis	(0.500, 0.000, 0.000)	2.135	0.000	(1.253, 0.000, 0.000)
single boosted source	(0.500, 0.000, 0.000)	1.067	3.561	(0.575, 0.000, 0.172)
asymmetric collision	(0.500, 0.000, 0.000)	1.921	0.712	(1.124, 0.000, 0.061)

The central symmetric case is intentionally a null test:

$$\mathbf{E}(0) = 0, \quad \mathbf{B}(0) = 0, \quad \Delta \mathbf{P} = 0.$$

The off-axis symmetric case checks the transverse electric kick while the two magnetic fields cancel. The single-source and asymmetric cases give nonzero magnetic fields and therefore exercise the full $(\mathbf{E}, \mathbf{B})$ Boris path.

3.3 Back-to-Back Breit-Wheeler Pair Propagation

The Python prototype also contains a two-particle witness test intended to be a minimal model for a low-energy Breit-Wheeler pair born at the interaction point. The electron and positron are initialized at the same lab event,

$$\mathbf{r}_{e^-}(0) = \mathbf{r}_{e^+}(0) = \mathbf{0},$$

with equal kinetic energy

$$K_{e^-} = K_{e^+} = 313 \text{ keV}.$$

The corresponding normalized momentum magnitude is

$$|\mathbf{P}| = \sqrt{\left(1 + \frac{K}{m_e c^2}\right)^2 - 1} = 1.26500561229, \quad \beta = 0.784487091427.$$

For the default test direction $\hat{\mathbf{u}} = \hat{\mathbf{x}}$,

$$\mathbf{P}_{e^-}(0) = |\mathbf{P}|\hat{\mathbf{u}}, \quad \mathbf{P}_{e^+}(0) = -|\mathbf{P}|\hat{\mathbf{u}}.$$

Three trajectories are integrated with identical initial conditions and timestep:

- **symmetric_collision**: the pair samples the Lorentz-transformed fields of two equal boosted Gaussian source bunches;
- **asymmetric_collision**: the pair samples the same left source and a right source with 80% of the charge;
- **no_fields**: the same pair is propagated freely with $\mathbf{E} = \mathbf{B} = 0$.

All cases are written into the same trajectory CSV with a **scenario** column.

For the physical-charge illustration

$$Q = -2.002721 \text{ nC}, \quad \Delta t = 0.005 \text{ ps}, \quad T = 50 \text{ ps},$$

the exact creation point is a symmetry null:

$$\mathbf{E}(0, 0) = \mathbf{0}, \quad \mathbf{B}(0, 0) = \mathbf{0}.$$

In the symmetric collision with the default pure transverse launch, both particles remain in the $z = 0$ symmetry plane, so the magnetic fields from the two equal and opposite source bunches cancel along the trajectory. The asymmetric collision breaks this cancellation and samples a nonzero magnetic field even for the same initial pair direction. The generated kinematic summary is:

case	particle	$\mathbf{r}_f$ [mm]	$\mathbf{P}_f$	K_f [keV]	$\Delta \mathbf{r}$ [μm]	ΔK [eV]
symmetric collision	e^-	(11.759488, 0.000000, 0.000000)	(1.265096, 0.000000, 0.000000)	313.036	(0.323, 0.000, 0.000)	36.248
symmetric collision	e^+	(-11.758843, 0.000000, 0.000000)	(-1.264915, 0.000000, 0.000000)	312.964	(0.323, 0.000, 0.000)	-36.246
asymmetric collision	e^-	(11.759456, 0.000000, -0.000041)	(1.265087, 0.000000, -0.000004)	313.033	(0.290, 0.000, -0.041)	32.623
asymmetric collision	e^+	(-11.758875, 0.000000, 0.000041)	(-1.264924, 0.000000, 0.000004)	312.967	(0.290, 0.000, 0.041)	-32.622

4 Implementation Use in OPALX

The recommended implementation sequence is:

1. Keep the Python evaluator as the reference for exact one-witness values.
2. Add an OPALX unit or regression test with one prescribed source and one witness particle.
3. First compare the field sampled by OPALX against the Python $(\mathbf{E}, \mathbf{B})$ value at the same event.
4. Then compare the witness momentum after one kick against the Python Boris update.
5. Only after this single-particle manufactured test passes should the implementation be generalized to many witnesses and mesh interpolation.

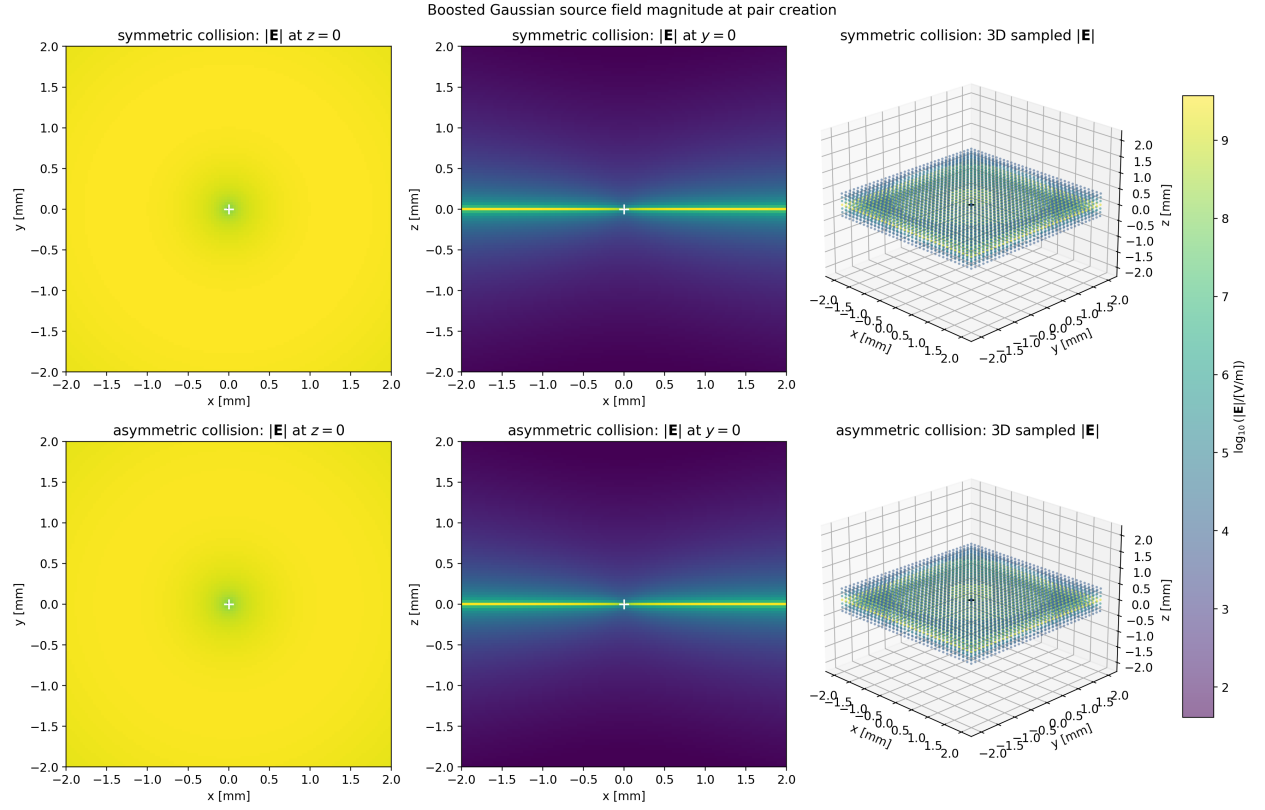


Figure 1: Magnitude of the manufactured boosted-Gaussian electric field at pair creation. The white marker is the interaction point.

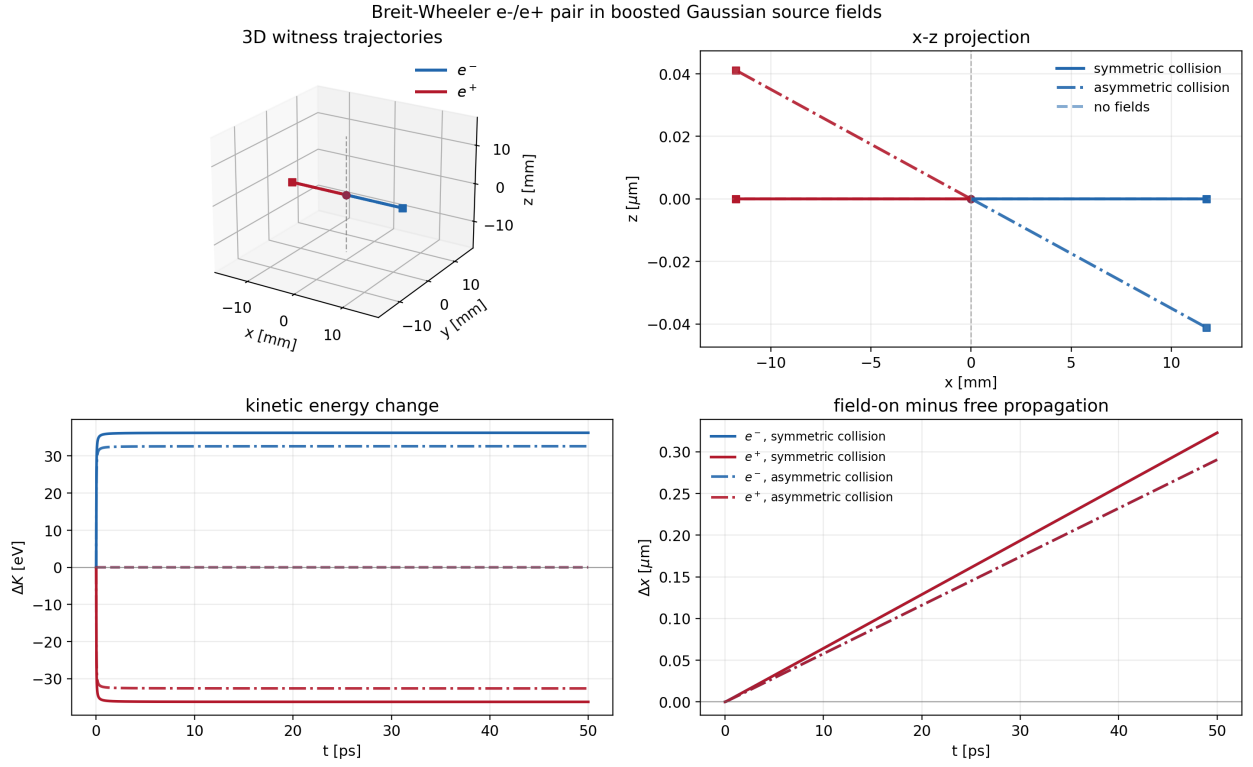


Figure 2: Back-to-back e^-/e^+ propagation with and without the colliding source fields. The lower-right panel shows the field-on trajectory minus the free trajectory.